Computer Vision-Based Construction Progress Monitoring

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Abstract

Automating the process of construction progress monitoring through computer vision can enable effective control of projects. Systematic classification of available methods and technologies is necessary to structure this complex, multi-stage process. Using the PRISMA framework, relevant studies in the area were identified. The various concepts, tools, technologies, and algorithms reported by these studies were iteratively categorised, developing an integrated process framework for Computer-Vision-Based Construction Progress Monitoring (CV-CPM). This framework comprises: data acquisition and 3D-reconstruction, as-built modelling, and progress assessment. Each stage is discussed in detail, positioning key studies, and concurrently comparing the methods used therein. The four levels of progress monitoring are defined and found to strongly influence all stages of the framework. The need for benchmarking CV-CPM pipelines and components are discussed, and potential research questions within each stage are identified. The relevance of CV-CPM to support emerging areas such as Digital Twin is also discussed.

Keywords

Progress monitoring, Computer vision, Automated construction, Data acquisition, 3D reconstruction, As-built modelling, Point cloud, Scan to BIM, Literature review, Digital Twin

Contents

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1. Introduction

Monitoring the progress of construction projects is crucial, as it provides vital inputs for managers to make timely and informed decisions. Improper progress monitoring leads to a loss of control of the project, resulting in time and cost overruns. Traditional progress-monitoring methods require manual data entry, which proves to be tedious, time-consuming, and prone to human error [1]. Therefore, its effectiveness can be improved through automation.

Technologies investigated for their applicability in automated progress monitoring in construction include RFID, barcodes and QR codes, laser scanning, photogrammetry, videogrammetry, range imaging (RGB-D), web-based CCTV, and structural sensing [2]. Among these, the impact of vision-based technologies was initially limited due to the sophistication of the data-acquisition devices and the high computing power required for processing. However, with the increasing ubiquity of both devices and high-performance computing, it has become feasible to implement vision-based technologies in applications that automate construction processes. Computer vision enables computers to derive numeric information from digital images, videos, depth images and 3D point clouds, process the information, and take action. Sub-domains of computer vision relevant to this paper include scene reconstruction, 3D pose estimation, motion estimation, object detection, object recognition and labelling, learning, and 3D scene modelling.

Vision-based technologies have been studied on a wide range of construction applications, such as workforce tracking, resource tracking, condition assessment, quality inspections, safety management, and automated layout generation [3]. Several studies on vision-based construction progress monitoring have also been reported. However, the processes for effective progress monitoring are diverse, intricate, and complex. In addition, there are many concepts and technologies that can be applied to automate each stage of the process. An integrated framework to characterise and categorise the current and future studies in this area will enable systematic investigation and documentation in this domain.

Review papers in this domain have focused on specific aspects of the progress monitoring process. Omar et al. have presented a review of data acquisition technologies [2], while Ma et al. [4] have comprehensively studied 3D reconstruction techniques. Ekanayake et al. [5] and Patel et al. [6] have recently presented a bibliometric analysis of the literature and highlighted specific challenges to be addressed. However, there are several stages in the process, and a broad study of both literature and practice has resulted in identifying the following key stages: (i) data acquisition and 3D reconstruction, (ii) generation of as-built models, and (iii) monitoring progress. Currently, there is no integrated framework that addresses the details of the processes of Computer Vision-Based Construction Progress Monitoring (CV-CPM), from data acquisition to progress estimation. Therefore, to address this gap, the following objectives are identified:

1. Develop an integrated framework that captures the process requirements of vision-based construction progress monitoring and enables the characterisation and categorisation of current and future work in the area.
2. Utilise the framework to position and compare the various concepts and tools adopted by published research studies.
3. Identify areas and strategies for future work in the area.

As a first step towards these objectives, relevant articles were selected through the broader steps defined in the PRISMA framework. These include identification, screening, eligibility, and inclusion. An extensive keyword search in the Scopus and Web of Science (WoS) reputable scientific databases was conducted. The keywords used were “progress monitoring in construction,” “construction progress monitoring,” “progress monitoring,” “automated progress monitoring,” “computer vision in construction progress monitoring,” and “vision-based progress monitoring in construction.” One hundred eighty-two articles were retrieved from the keyword search. Primary screening of the articles was carried out by reading their abstracts. In order to capture relevant developments in this rapidly evolving area, only key articles appearing after 2014 were defined as eligible for meta-analysis. Other articles which address specific parts of the framework were also referred to and positioned appropriately in the manuscript to support the framework. For an article to qualify as a key article for review, the following two criteria were set:

- “The article should present an applied computer vision-based progress monitoring pipeline by demonstrating it with a case study or experimental results.”
- “Pipelines addressing both indoor and outdoor progress monitoring, as well as pipelines ending at different stages of a type of progress estimation, were considered (i.e., only visualisation as well as visualisation with quantification).”

Twenty-four key articles were selected and included for meta-analysis to formulate the framework. Among other articles referred for specific tools and techniques, fifty-four articles are from the civil engineering domain, and eight are from the computer science domain.

The key articles were systematically reviewed, analysed and categorised based on the process they addressed and the concepts they used, following the methodology defined by [7]. The framework evolved as more studies were characterised, categorised, and positioned within it, and the application contexts of concepts and technologies used were studied and compared. After the reviewed works were positioned in the framework, the processes, concepts, and technologies that need to be further developed/explored were identified.

The organisation of this paper is as follows. Section 2 characterises and classifies the key studies in this area, based on the three macro-stages of the CV-CPM process. Section 3 expands the three macro-stages into a detailed process framework, and for each process in the pipeline, the relevant concepts, technologies, and tools used are compared and discussed. Section 4 discusses a strategy for a systematic approach in this domain and identifies the areas for further work. The conclusions of the study are presented in Section 5.

2. Overview of vision-based monitoring of construction progress

This section presents the macro-level stages proposed for vision-based progress monitoring in construction and references the key progress monitoring processes within these stages. Each of these processes is also referred to as a pipeline. Figure 1 shows the six macro-level processes that constitute vision-based progress monitoring.

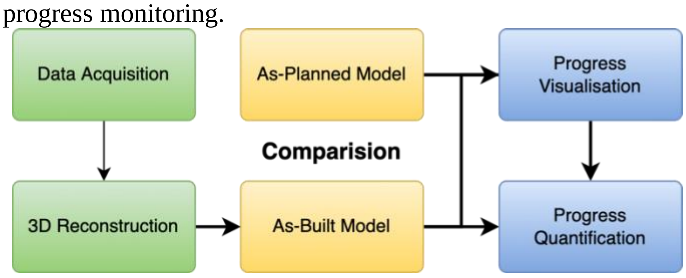

Figure 1 Macro-level conceptualisation of vision-based progress monitoring

Table 1 Review of pipelines for vision-based progress monitoring

Papers	Data Acquisition			3D Reconstruction	As-Built Output	Progress Monitoring				Identified Elements	Operational State	Key Focus	LPM
	Technology	Method	Indoor/ Outdoor			Visualization	Comparison	Quantification	Schedule Updating				
[45]	Laser Scanning and Photogrammetry	Manual	Indoor	SLAM	3D Mesh	Extended Reality XR (VR+AR)	Visualization	N	N	Target Cuboidal Objects	N	Immersive VR (iVR)–based visualization of progress	L-2 Visualization and comparison
[13]	RGBD Camera	Manual	Both	Virtual photogrammetry	Segmented Images	Interactive VRE	Object Detection	N	N	Building Elements (Walls, floor, beams etc.,)	Y	CNN-based object detection and use of gaming engine	
[57]	Digital Images	UAV	Outdoor	Commercial Software	Segmented Image with Point Cloud	3D Viewer	Image and Point cloud-based Object Detection	N	N	Columns	Y	Computer vision-based element detection using images, camera poses, as-planned-BIM, precedence relationships	
[44]	Laser Scanning or Photogrammetry	Manual	Both	Commercial Software	Point Cloud	VRE	Commercial Software	N	N	Not specified	N	Gaming engine utility for visualization of as-built and as-planned data	
[38]	Photogrammetry / LS	UAV	Both	SfM & MVS	Point Cloud	3D Viewer	Thresholding	N	N	Construction Materials	Y	Uses Geometry and Appearance based reasoning	
[50]	Photogrammetry	Manual	Outdoor	VSfM / SGM	Point Cloud	3D Viewer	Projection Thresholding	N	N	Columns	N	Uses Graph Theory and precedence relationships for Progress	
[34]	Photogrammetry / LS	Manual	Both	SfM & MVS	BIM	3D Viewer	Ontology based Deviations	N	N	Not specified	Y	Uses Construction Sequence Ontology	
[28]	Photogrammetry	UAV	Outdoor	SfM	3D Mesh	AR viewer	Interactive AR	N	N	Building Façade	N	AR-based Progress monitoring using mobile aerial 3D reconstruction and visualization	
[58]	Photogrammetry (Unordered Images)	Manual	Both	Model-assisted SfM	Point Cloud	ConstructAide GUI	Visual Colour Coded	N	N	Not specified	N	Uses user assisted SfM method	
[9]	Photogrammetry (Unordered Images)	Manual	Both	SfM & MVS	BIM	web-based viewer	Material Recognition	N	N	Construction Materials	Y	Appearance-based material classification using LM & HSI features	
[59]	Videography	Fixed	Both	NA	Object in Image	Image viewer	MATLAB Image Processing	Y	N	Columns, beams, block masonry	N	MATLAB-based element cropping and quantity estimation	L-3 Quantification
[60]	Laser Scanning and Photogrammetry	UAV	Both	Manual Modelling	Mesh Model/ 3D BIM	3D Viewer	Overlapping & BoQ	Y	N	Structural Components	N	Hybrid data collection using drone-based photogrammetry and Laser Scanning	

Papers	Data Acquisition			3D Reconstruction	As-Built Output	Progress Monitoring				Identified Elements	Operational State	Key Focus	LPM
	Technology	Method	Indoor/ Outdoor			Visualization	Comparison	Quantification	Schedule Updating				
[61]	Video Camera	Fixed	Outdoor	NA	Object Detected & Segmented Images	3D Viewer	Object Detection	Y	N	Precast Walls	N	Object detection, segmentation and multiple objects tracking from images to detect precast walls.	L-3 Quantification
[12]	RGBD Camera	Manual	Indoor	Commercial Software	3D Mesh	Mixed Reality	Rays Thresholding	Y	N	big regular-shaped objects	N	Mixed Reality–based progress monitoring	L-3 Quantification
[31]	Photogrammetry	Manual	Both	SfM & MVS with Coded Targets	BIM	3D Viewer	BoQ	N	N	Walls	N	Enhanced method for 3D reconstruction using coded targets	L-3 Quantification
[26]	Laser Scanning	Manual	Both	Commercial Software	Point Cloud	3D Viewer	Overlapping Distance Threshold	Y	N	Rectangular Columns	N	Extraction of columns using geometric and relationship-based reasoning	L-3 Quantification
[48]	NA	NA	Both	NA	Point Cloud	3D Viewer	Thresholding	Y	N	Walls	N	Percentage of completion of in-situ cast concrete walls	L-3 Quantification
[10]	Videogrammetry	UAS	Outdoor	VSfM / CMPMVS	3D BIM	3D Viewer	Thresholding	Y	N	Building Façade	N	Uses UAS images, low-cost photogrammetry, and GIS	L-3 Quantification
[62]	Photogrammetry	Manual	Both	Commercial Software	Point Cloud	3D Viewer	Object Detection	Y	N	Masonry Walls	Y	Use of feature engineering for object detection and data classification	L-3 Quantification
[11]	Laser Scanning	Manual	Both	Commercial Software	BIM	3D Viewer	Object Recognition	Y	N	MEP Components	N	Use of Hough Transform to detect parametric features with Scan Vs BIM	L-3 Quantification
[8]	Photogrammetry (Unordered Images)	Manual	Both	SfM & MVS	Point Cloud	D4AR Viewer	Voxel Occupancy	Y	N	Columns, Foundation Walls	N	Use of machine-learning scheme built upon a Bayesian probabilistic model	L-3 Quantification
[63]	Photogrammetry (Unordered Images)	Manual	Both	SfM & MVS	Point Cloud	web-based viewer	Material Recognition	Y	N	Construction Materials	Y	Appearance-based material classification	L-3 Quantification
[47]	Photogrammetry	Manual	Outdoor	Commercial Software	Point Cloud	NA	Enclosed Volume Calculation	Y	Y	Columns	N	Photogrammetry-based monitoring with automated notification system and schedule update	L-4 Schedule Update with notifications
[14]	RGBD Camera	Manual	Both	Commercial Software	4D BIM	3D Viewer	Commercial Project Management Software	N	Y	Not specified	N	Continuous monitoring using helmet-mounted scanners	L-4 Schedule Update with notifications

These processes are grouped into three stages, as follows:

i. Acquisition of as-built data and conversion into a point cloud using 3D reconstruction techniques.
ii. Generation of the as-built model from the point cloud (with reference to the as-planned model, if available)
iii. Comparison of the as-built and as-planned models to assess progress by visualisation and/or by quantifying the work done.

All studies on vision-based progress monitoring focus on one or more of these stages. Table 1, a key output of this study, was developed through a detailed review of the published works on vision-based construction progress monitoring. The articles were identified through the systematic process described in the introduction. The categories shown in Table 1 were refined iteratively based on objectives, solutions and concepts addressed by the reviewed papers.

The studies referred to in Table 1 are arranged based on the category of progress monitoring, as indicated in the last column. The first column contains the reference to the study, and the subsequent columns contain the categories and sub-categories based on the macro-level model. The cells in Table 1 identify the technology, concept, and algorithm explored in each category and sub-category. It can be observed in Table 1 that for each category, there is a wide range of alternatives for all process components, construction elements, and the key foci that the studies have explored, as discussed briefly in the following paragraphs.

Data Acquisition and 3D Reconstruction: This section of the table documents the data acquisition technologies, type of mounting, and usage environment (indoor/outdoor) that the studies use. Depending on the input type, the data undergoes 3D reconstruction using various commercial software, as shown in Table 1. The workflow from data acquisition to 3D reconstruction is discussed in detail in subsection 3.1 of this paper.

As-built Modelling: As shown in Table 1, there are varied output data types that are generated by the pipelines. The generation of these as-built models requires specific as-built modelling techniques, which are classified and discussed in detail in subsection 3.2.

Progress Monitoring: Table 1 also indicates the level of progress monitoring (LPM) carried out, judging by the types of platforms used to implement progress estimation. This study classifies these into four levels. These are: (L-1) only visualising the as-built model, (L-2) visualisation incorporating a comparison with the as-planned model, (L-3) quantification of progress, and (L-4) quantification with schedule update and warning notifications.

For progress visualisation, different immersive and non-immersive environments have been used. For quantification, numerous approaches have been used based on the availability of an as-planned model. The details are discussed in subsection 3.3.

In addition to the three process components—data acquisition and 3D reconstruction, as-built modelling, and progress monitoring—Table 1 also characterises the pipelines on two additional parameters. First, the types of elements being worked on, and second, whether the structure's current operational state can be recognised (e.g., a wall's stage can be brickwork- completed, plastered, or painted).

As inferred from Table 1, numerous combinations of tools, concepts, and algorithms need to be evaluated to arrive at the appropriate approach for a specific progress monitoring need. Though all the combinations may not be technically viable, a significantly large set of options needs to be experimentally explored and documented to ensure systematic progress in this domain. Therefore, an integrated framework encompassing all the processes will support the systematic study.

In the next section, the three macro-level steps are expanded to formulate the detailed framework wherein the concepts/processes/technologies mentioned in these studies are characterised, positioned, and discussed.

3. Integrated framework for Computer Vision-Based Construction Progress Monitoring (CV-CPM)

The integrated framework for Computer Vision-based Construction Progress Monitoring is shown in Figure 2. This framework was derived after evaluating the existing literature studies in detail using the macro-level framework lens described in the previous section and the methodology defined in [7]. This section explores the micro-level details of each of the three main components. In addition to the basic concepts and references to the work done in this area, Table 2 summarises the key contributions of this paper for the micro-level stages of the framework with reference to other review papers.

The following three sub-sections are on the various processes, algorithms, methods, and technologies that could be applied to the relevant subsections of the framework.

Table 2 Summary of stage-wise specific contributions

Stage	Published review papers		Detailed contribution of this paper
	Reference	Their contribution	
Data Acquisition	[2]	Categorised, listed, and compared various data acquisition technologies and studies in detail.	• Identified and proposed factors and ratings that can be used to guide selection of sensors-mount combinations for data acquisition.
3D Reconstruction	[4]	Presented and overall pipeline for 3D reconstruction using SfM. Categorised, listed, and compared techniques for SfM technologies and studied in detail.	• Comparison SLAM and SfM technologies for construction progress monitoring.
As-Built Modelling	[32]	Presented an overview of the heuristic-based modelling process and classification of categories.	• Defined types of as-built models based on existing work and proposed guidelines for selecting as-built model type for progress monitoring based on project requirements. • Detailed review of as-built modelling approaches and rationale for using a hybrid approach.
Progress Visualization	NA	NA	• Categorised progress monitoring into four levels and associates these levels as the requirements driver for deciding on options in the upstream stages of process.
Progress Quantification	NA	NA	

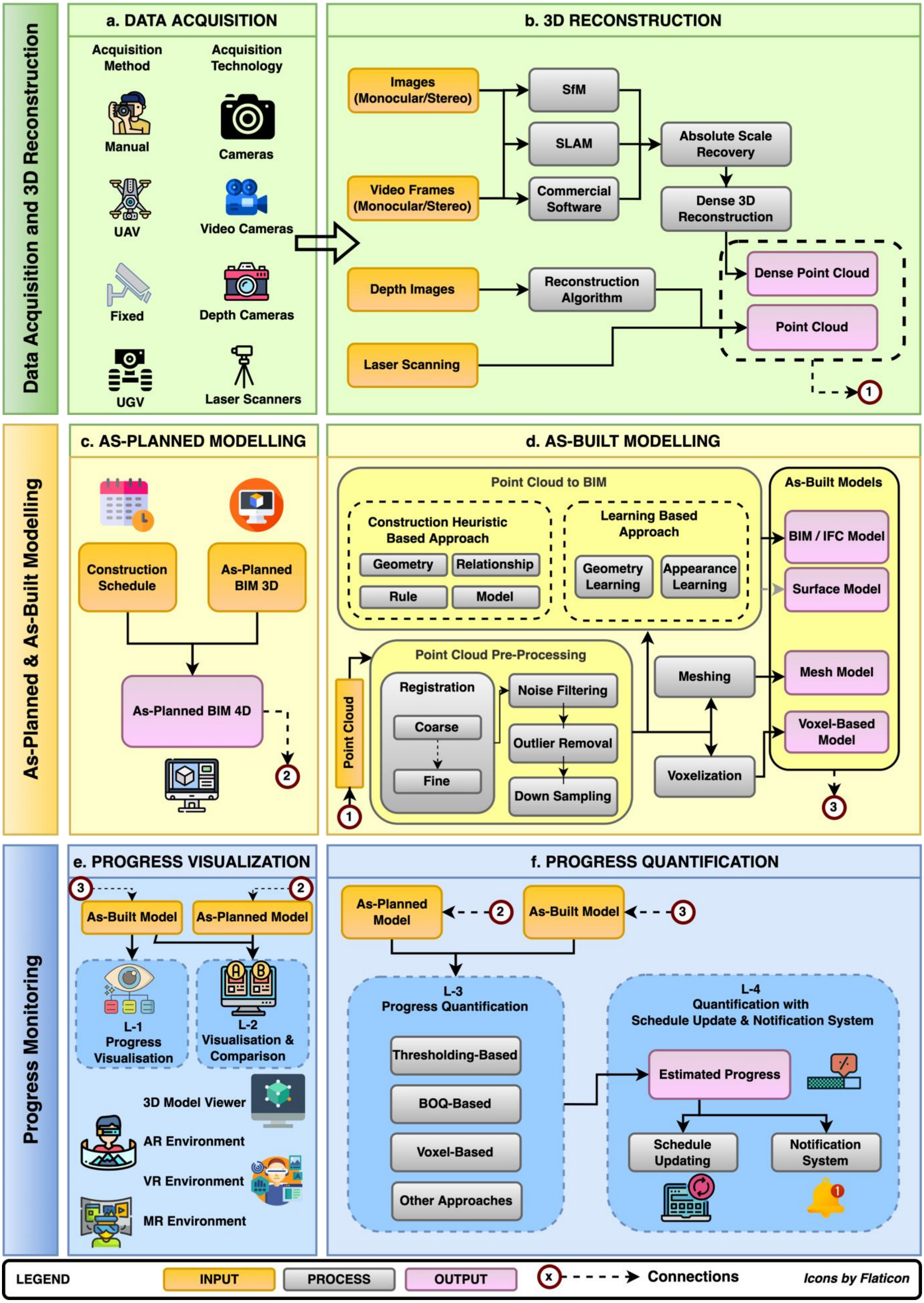

Figure 2 Integrated framework for computer vision-based construction progress monitoring (CV-CPM)

3.1. Data acquisition and 3D reconstruction

This section addresses the data acquisition (Figure 2 (a)) and 3D reconstruction (Figure 2 (b)) components of the framework. The data acquisition depends on the acquisition technology as well as the sensor mounting method selected. There are various steps involved in 3D reconstruction algorithms. The decision as to which technologies to use is a multi-faceted issue. Defining these based on the characteristics of a construction project and other factors is the primary contribution of this section.

3.1.1. Data acquisition: technologies and methods

Data acquisition depends on selecting the technology and the sensor mounting method (Figure 2 (a)). As seen in Table 1, several combinations of sensing technologies and sensor mounting methods have been explored. Digital cameras [8][9], video cameras [10], laser scanners [11], and range imaging (or RGB-D cameras) [12][13][14] are the vision-based technologies used for collecting data for progress monitoring. They generate inputs to the framework in the form of image frames or point clouds. The sensor mounting method for data acquisition can be in the form of fixed devices, handheld devices, robotic systems mounted on unmanned ground vehicles (UGV) [15], unmanned aerial vehicles (UAV) [10], or a combination of these systems [16].

A systematic review can be found in [2] on technologies used for progress tracking in construction. The study finds technology selection a "multi-faceted issue" which depends on "required degree of accuracy, project size, level of automation, and the ultimate purpose of progress tracking".

Figure 3 shows a dual matrix illustrating the data acquisition technology and method selection criteria. The technologies and methods are listed on the concentric rings in their respective matrix, where colours and icons are added for easier comprehension. The key factors that govern their selection have been identified and are positioned along the outer ring. Eight key factors for the sensor mounting method include statutory clearance, cost of mounting equipment, range of operation, preferred use case, operator training, navigation, manoeuvring speed, and accessibility. In addition, twelve key factors for data acquisition technology have been identified. They include the level of automation for data capture, real-time data availability, range of equipment operation, spatial resolution, spatial accuracy, adequate lightning requirement, user training requirement, time for data capture, preparations for data capture,

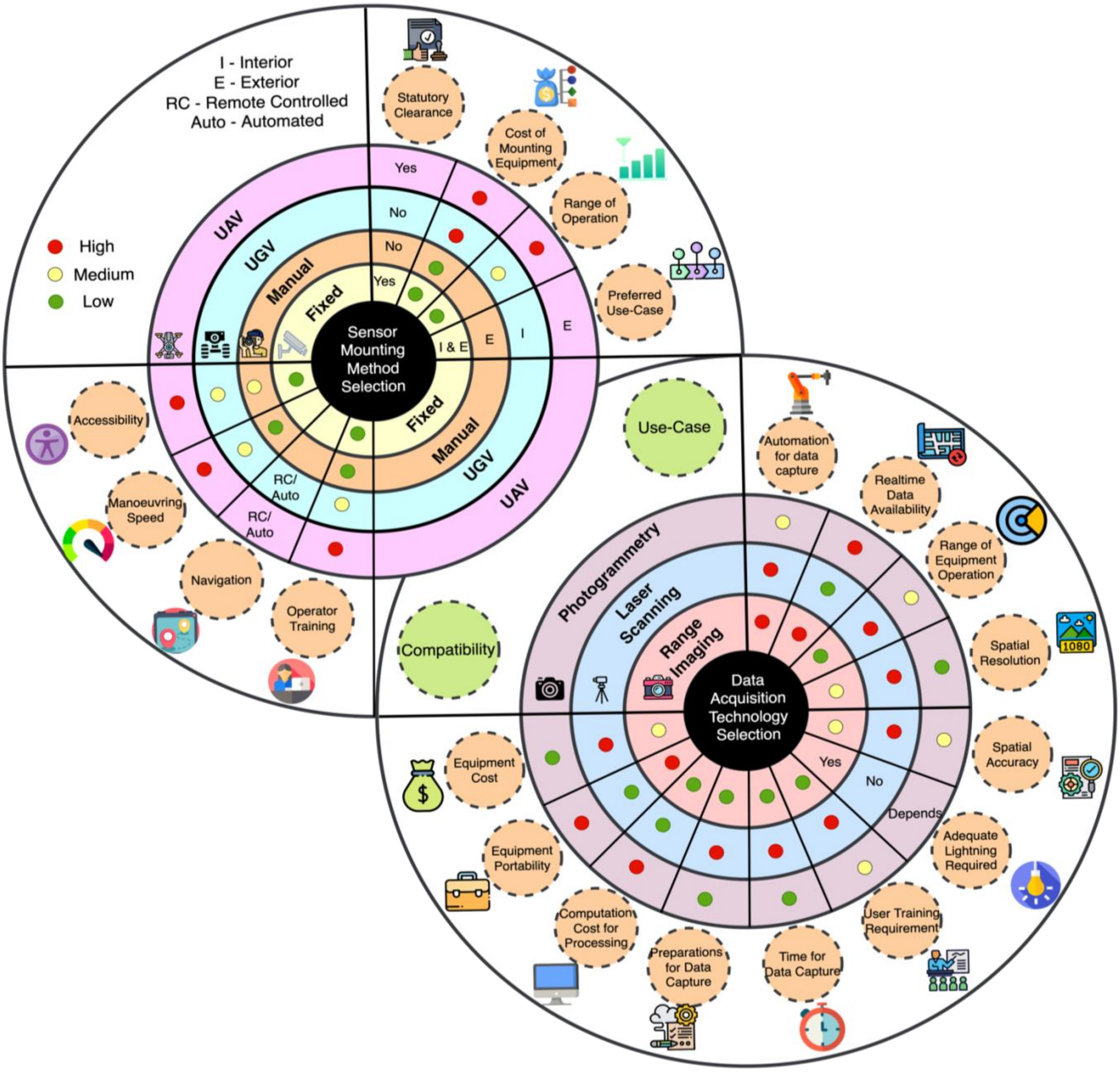

Figure 3 Dual matrix for sensor mounting method and data acquisition technology selection

computation cost for processing, equipment portability, and equipment cost.

This matrix is constructed based on the review and classification of literature. The relative comparison indicator for technology versus sensor mounting method is indicated along the radial direction. The low, medium, and high ratings are shown using green, yellow, and red colour codes, respectively. The proposed rating is a preliminary step towards assisting in the selection of a combination. As the proposed ratings are based on the authors' perspective, experimental studies are required to make them more objective.

3.1.2. 3D reconstruction

Following the image-based input from the previous step, the next step is to generate a point cloud model from a series of algorithms for 3D reconstruction (Figure 2 (b)). As observed from Table 1, most photogrammetry-based progress monitoring pipelines use commercially available software to generate a 3D point cloud from optical camera images or depth images.

Optical camera images do not contain depth information. SfM (Structure-from-Motion) and SLAM (Simultaneous Localisation and Mapping) are the two conceptual approaches to add depth information for sparse 3D reconstruction.

Depth images are captured using RGB-D cameras, for example, Microsoft Kinect. They have depth information (XYZ coordinates) along with colour information (RGB values) on a per-pixel basis. This information is used for mapping and performing dense 3D reconstruction using intrinsic and extrinsic camera parameters [17]. The output from a laser scanner is a 3D point cloud, and hence does not require 3D reconstruction.

The work in [4] focuses on generating point clouds from monocular and stereo images, and video frames using SfM for photogrammetry and its corresponding algorithms for 3D reconstruction. In our paper, only the basic concept of SfM is presented, and a detailed comparison with SLAM is made to facilitate decision making for 3D reconstruction in construction.

SfM (Visual SFM [18] or Open SfM [19]) is a technique consisting of a combination of algorithms for photogrammetric 3D reconstruction from numerous image frames (Figure 2 (b). It is an offline approach for estimating the scene's camera motion information. The process is to match all the images to each other, find the correspondences, and then delete mismatched images to obtain relative camera positions and the structure without any prior geometric or semantic information. The concept of SfM is based on stereoscopic photogrammetry, as stated in [20]. A detailed review of the 3D reconstruction by SfM and its algorithm can be obtained in [4] and [21].

SLAM [22] is a more general framework, in real-time, where a mapping system starts motion from an unknown location in an unknown environment, and during movement, simultaneously keeps track of its position with respect to the surrounding environment, building an incremental map. SLAM may depend on sensors like cameras, LiDAR, GPS, IMU, etc. Modern visual SLAM has gradually developed into a multi-feature, multi-sensor, and deep learning-based method that can be used for dynamic and haphazard construction environments.

As shown in Table 1, the use of Visual SLAM in progress estimation has been sparsely explored in construction. With the rapid advances in autonomous technologies for data acquisition (UAV/UGV based), SLAM-based reconstruction will become more relevant in construction.Figure 4 shows the comparison between SfM and SLAM to guide the selection process for 3D reconstruction applications for progress monitoring. The three key factors are discussed below in detail, namely:

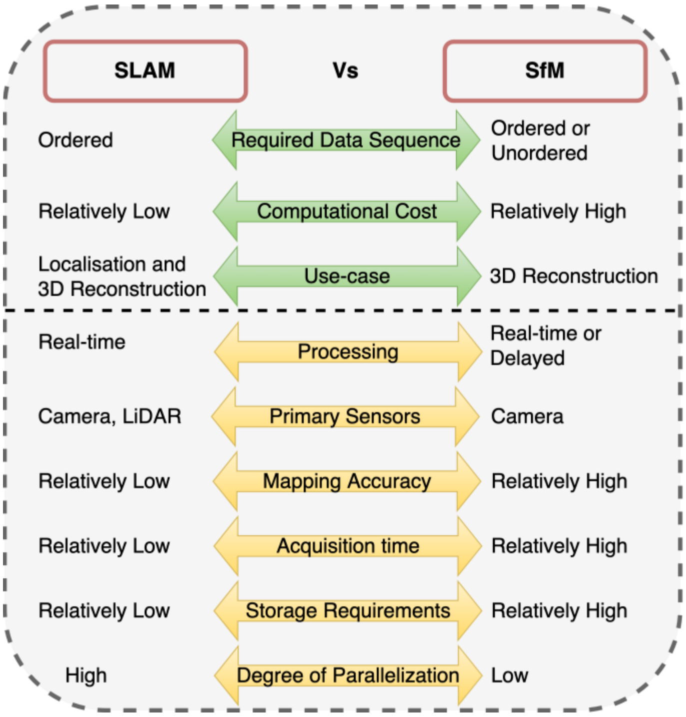

Figure 4 Comparison between SLAM and SfM for 3D reconstruction in construction

1. **Data sequence:** SfM is based on feature matching of image pairs is highly dependent on the quality and the sequence of the image frames obtained. It need not necessarily require ordered image frames. On the other hand, SLAM builds an incremental map in real-time and requires sequential frames to estimate the previous and next poses. Hence, construction sites collecting images from multiple location-aware sensors should be able to work on SfM. SLAM can be selected for sites requiring autonomous navigation of robots in volatile unmanned environments.
2. **Computational cost:** The process of SfM is computationally costly because of unordered data compared to SLAM, which works on ordered data. Bundle adjustment (non-linear optimisation) [23] in SLAM is applied only on the last "N keyframes", as opposed to the entire graph in SfM, to give a real-time performance in budget. In scenarios where the availability of parallel GPUs is not a constraint for computation, SfM will give better results than SLAM.
3. **Path-planning or Autonomy:** Pre-planning the robot surveillance paths for construction sites is tedious and complex. Planned paths are subjected to uncertainty on construction sites due to their dynamic nature. Both SfM and SLAM can be used if the sensor is mounted on a UAV or a UGV. SfM is predominantly used in known environments where surveillance paths can be planned. In contrast, SLAM is advantageous for autonomous applications in random locations and unknown environments.

Apart from these three key features, other sub-factors which can be considered are shown in Figure 4. In general, SLAM generates a sparse map compared to SfM and has limitations in loop closure, scaling, and drifting issues. However, SLAM can overcome SfM's drawback when dealing with a featureless or repetitive scene with potential false matches. Also, SfM has been tried on a larger scale, whereas SLAM can be used for small-scale tasks.

It is evident from the review that both technologies have their advantages and disadvantages, and selection should be based on project requirements and compatibility. There are multiple algorithms for SLAM [24] and SfM ([18],[19]), and when

Table 3 Sub-processes and associated techniques

Sub-Process	Techniques/ Algorithms	Existing Use in Progress Monitoring Pipelines	Other Works
Absolute Scale Recovery	Manual	-	[64]
	Pre-measured Object	[31][63]	
	Geo Registration	[28][10]	
Dense 3D Reconstruction	Multi-view Stereo MVS	[63][38]	[18][19] [25]
	Clustering Multi-View Stereo (CMVS)	[31]	
	Patch-based Multi-View Stereo (PMVS)	[31]	
	MVS with voxel coloring	[8][9]	
	SGM	[50]	
	Meshing using PSR	[10][28]	
Registration	**Coarse Registration**		[65][66] [25]
	Manual	[8][60][38]	
	Marker-Based	[28][26]	
	Sensor-Based	[28]	
	Feature-Based	[31][67]	
	Fine Registration		
	ICP	[28][68][45][50] [14][69][57]	
	Image Registration		
	Geo-referencing	[58][34][70][63] [21]	
Noise and Outlier Removal	Manual	[60]	[71][72]
	RANSAC+PCA-based	[26]	
	Tensor voting algorithm	[68]	
Down Sampling	Point-space strategy Algorithm	-	[73]

various types of embedded sensors are included, the options for selection increase. For example, the embedded camera sensor can be monocular, binocular or RGB-D. Based on the type of sensor, the core of the reconstruction algorithm changes, as well as the accuracy and the computation time. Therefore, numerous combinations of these variations increase the complexity of the framework and the resulting selection process.

3.1.3. Absolute scale recovery and dense 3D reconstruction

The output obtained by the image-based pipeline is primarily sparse and not to scale. Therefore, as shown in Figure 2 (b), it requires absolute scale recovery and dense 3D reconstruction. Laser scans do not require this. Table 3 shows the techniques and algorithms that can be used in the associated sub-processes.

The scale recovery sub-process determines the absolute scale of the generated sparse point cloud by comparing it with the local coordinates of the point in the sparse point cloud. Scale recovery has been implemented using, manual techniques, pre-measured objects, and geo-registration, as shown in Table 3. The first two techniques require manual measurements and feeding ground truth, whereas geo-registration requires automated sensing of camera parameters to apply the transformation.

The next sub-process, Dense 3D reconstruction, recovers the scene details. The algorithms used are multi-view stereo (MVS), clustering views for multi-view stereo (CMVS), patch-based multi-view stereo (PMVS), and semi-global matching (SGM). The open-source algorithms which, perform dense 3D reconstruction include VisualSfM [18], OpenSfM [19], and Bundler SfM [25].

A dense 3D reconstruction step is required if the reconstructed point cloud is sparse, typically in the case of reconstruction from monocular or stereoscopic images or video frames. The reconstruction output from depth images is already dense enough to be used directly and does not require dense 3D reconstruction.

Only a handful of CV-CPM pipelines have stated the method used for absolute scale recovery and dense 3D reconstruction; others have not explicitly mentioned it in the paper. The choices available for absolute scale recovery and dense 3D reconstruction add additional variables to our framework. Once a dense point cloud is generated, the next step is to pre-process it through registration, noise filtering, outlier removal and down sampling. These sub-processes and model generation are discussed in the next section.

3.2. Model generation

In this section, first, the characteristics of the as-planned model are discussed. All the steps in pre-processing and the process of generating the as-built model are then discussed in detail (Figure 2 (d)). The existing techniques for as-built model generation are categorised into heuristic-based and learning-based methods. The definition of these categories and the mapping of existing pipelines to these categories/sub-categories is the primary contribution of this section. This section also presents the trade-off between the levels of information the various as-built outputs provide for progress estimation and the computational complexity required to generate them.

3.2.1. As-Planned model

As-planned models may or may not be available for a particular facility or project. Generally, they are prepared in the design phase of the project. These can be 2D/3D CAD or 3D/4D BIM models. The approach to progress monitoring depends on the availability and type of the as-planned model, as it facilitates comparison with the as-built model.

3.2.2. As-Built model

The following sub-sections include the details for the various sub-processes and options for as-built modelling (Figure 2 (d)).

3.2.2.1. Point cloud pre-processing

Point cloud pre-processing is a crucial step towards as-built modelling to improve the point cloud quality. The level of automation in pre-processing has not been adequate, and manual intervention is still used to get the required quality. Table 3 outlines the critical sub-processes of point cloud pre-processing, which broadly consist of registration, noise filtering, outlier removal, and down-sampling, along with various techniques and algorithms used in the corresponding literature. The table also shows studies that have addressed fundamental aspects of these sub-processes.

Registration involves merging multiple partial point clouds obtained into a single file or registering an as-built point cloud over an as-planned BIM model for progress monitoring. There are two stages of registration, coarse and fine.

Coarse registration aligns a set of point clouds using correspondences between them. In contrast, the fine registration algorithm further matches multiple point clouds by estimating a rigid transform and minimising the distance between the corresponding matched points. The selection of these approaches depends on the type of data acquisition technology used and location capture, as registration depends on localising the data. While the manual and marker-based methods require iterative human efforts, the sensor and feature-based approaches are reported to be more automated. The Iterative Closest Point (ICP) algorithm is the most used in CV-CPM pipelines for fine registration. It searches for an optimum solution, and the convergence speed is directly proportional to the accuracy of coarse registration. For image-based methods, image registration over the BIM model is made using georeferencing.

Next, noise and outliers are unwanted points and are removed to improve the point cloud quality prior to further processing. Finally, due to the merging of point clouds, the overlapped regions become much denser, affecting processing efficiency. Hence down-sampling is performed to make the point cloud uniform. Relevant works which include these steps are shown in Table 3.

Although pre-processing of data has been presented in some progress monitoring pipelines, there is no detailed exploration of the available techniques. There is a need for a comprehensive study and further research to customise pre-processing methods based on the input pipeline's requirement. There is also a need to explore the other techniques mentioned in Table 3 for better results using the CV-CPM framework.

3.2.2.2. Types of as-built models:

Existing pipelines have used several combinations of as-built and as-planned model comparisons for progress estimation (Table 1). Based on the algorithm and method used, the as-built output can be a pre-processed point cloud [26], voxel model [27], mesh model [12][28], surface model [29][30], or IFC/BIM models [31][14][11] (Figure 2 (d)).

The level of progress monitoring proposed for a project plays a vital role in deciding the type of as-built model to be generated. Table 4 compares computational requirements and quality of output for the different levels of progress monitoring with the corresponding types of as-built models. Here, green represents a performance improvement, and red indicates a decrement in performance.

It can be seen from Table 4 that for L-1 or L-2 based monitoring, voxel-based, mesh-based, or point cloud models are recommended to be used because of low computational requirements. Even if the point cloud quality is low, it is adequate for visualisation. While visual output quality is better for IFC-based or surface models, it is not recommended for progress visualisation due to very high computing requirements.

For L-3 or L-4 based monitoring, while it will take higher computation to generate an IFC or surface-based model initially, the computation required for progress quantification will be substantially lower with higher accuracy; hence it is recommended. On the other hand, the mesh model, voxel model, and point clouds are easy to generate but will take higher computation to perform the comparison and result in low accuracy. Mesh-based and voxel-based modelling are explored domains that are easier to develop using established algorithms and software and therefore were briefly introduced.

Hence, this trade-off of required computation, accuracy, speed, and application case should be considered while designing the progress monitoring pipeline for a specific usage requirement. The relative comparison provided here is a preliminary step based on the authors' perspective. To arrive at more objective ratings, the parameters must be assessed quantitatively (for computation and accuracy) and qualitatively (for visual output quality). Further experimentation is required to make these assessments.

Progress visualisation (L-1 & L-2) is typically being implemented at construction sites. However, to take critical decisions, quantifying progress is essential. IFC-based as-built models can provide direct and accurate progress quantification (L-3) and facilitate schedule updating and notifications generation (L-4). Therefore, existing studies have explored various techniques to achieve point-cloud to IFC/BIM model conversion. The following sub-section discusses and classifies these approaches in detail.

3.2.2.3. Point Cloud to BIM

This section focuses on as-built modelling methods resulting in IFC classes (Figure 2 (d)). Only two existing studies have classified point cloud-to-BIM approaches [32][33]. One has classified them as local and auxiliary heuristics based on as-planned BIM involvement [32], whereas the other extends the classification based on geometry, rule, model, and feature-based techniques [33].

Table 5 summarises the approaches taken by the referenced studies to explore the point cloud-to-IFC/BIM conversion. This study broadly classifies these approaches as heuristics-based and learning-based and into additional subclasses, extending the classification by [32] and [33].

The use of heuristics has been based on geometry, rule-based, relationship-based, and model-based constraints. Learning-based techniques can be generally classified as geometric and appearance learning. The following section discusses and analyses the two approaches of element recognition in detail.

A. Heuristics-based approach

Heuristics-based approaches (Figure 2 (d)) use pre-coded domain knowledge to identify elements from as-built point clouds. Table 6 shows the comprehensive sub-classification of identified heuristics with their definitions as formulated by the authors and the context of their applications in the literature.

Table 4 Relative computational requirements and output accuracy for the level of progress monitoring, based on the type of as-built models

Type of as-built models	**Level of progress monitoring**			
	L-1 and L-2		**L-3 and L-4**	
	Progress Visualization		**Progress Quantification**	
	Computation for model generation	**Quality of visual output**	**Computation for progress quantification**	**Accuracy of Qty. take-offs**
IFC/BIM	Very High	Very High	Very Low	Very High
Surface Model	High	High	Low	High
Mesh Model	Low	Medium	High	Medium
Voxel Model	Low	Medium	High	Low
Point Cloud	Very Low	Low	Very High	Very Low

Following are a few insights from Table 6 for the four types of heuristics:

1. **Geometrical constraints** have been applied based on shape, dimensions, point density, and associative geometry. They require only basic geometric information as inputs; therefore, they are widely utilised. The challenge lies in retrieving and feeding this information for all the elements, as construction sites may have unconventional elements.
2. **Rule-based constraints** have been applied based on the direction of the surface normal, orientation, principal axis direction and other rules. Apart from a few complex elements, most elements can be identified by applying these rules.
3. **Relationship-based constraints:** The construction sequence is of utmost importance, as it can accurately define the objects' dependencies from schedules, enabling better element detection for progress monitoring. These time-based relationships can be helpful while there is limited visibility of the scene or occlusions. Space-based constraints have been used in the literature using ontological relations [34] and shape grammar [35], but they are challenging to formulate and apply in practice.
4. **Model-based constraints:** Model-based element retrieval uses reverse engineering to recognise the elements. The point cloud is registered over the as-planned BIM model, and the elements are identified based on overlap occupancy. Although the method results in high accuracy, its use is limited because it requires a detailed as-planned model. It has specific point cloud quality and reference BIM model requirements [36]. The approach also faces computational challenges in automatic registration as the acquired point cloud data is voluminous.

It can be summarised that heuristics-based techniques usually require a significant amount of prior information. Generation of this information requires domain knowledge about construction and the type of elements being recognised. A recommended solution is to define a set of metaheuristics that generally applies to any construction dataset as a subset of all the heuristics; using them will make element identification viable with less detailed information requirements.

Though using some heuristics makes element identification easier, a pure heuristic-based approach for CV-CPM can be unreliable and challenging to be implemented.

B. Learning-based approach

Learning-based techniques (Figure 2 (d), Table 5) are used on point clouds for 3D shape classification, object detection, and segmentation. They are based on training features using a neural network and computed weights to predict object or material classes. Two types of learning approaches have been used:

1. **Geometric Learning** extracts geometrical features using descriptors and feeds neural networks on statistics like distance, area, and angles. The geometric features are used to identify the element's physical presence.
2. **Appearance Learning** extracts features like HSI (Hue-Saturation-Intensity) colour values, material reflectance, and surface roughness as features. The appearance-based features describe its operational state.

As observed from Table 5, only a few studies have detected elements for as-built modelling using learning-based techniques in the past. Most of these methods detect geometric features and do not recognise the elements' operational state. A few studies recognised the operational state by using image processing techniques [37][38] and learning appearance-based features [39].

For progress monitoring applications, learning-based approaches have been predominantly used on 2D image data, and there are very few studies involving 3D point cloud learning. Recent advances in machine learning and deep neural networks have motivated researchers to explore this area. However, their application in point cloud-based element recognition for construction has been limited because of challenges such as:

1. 3D point cloud data are unstructured, large, and with varying point densities, making 3D feature detection computing-intensive and time-consuming.
2. Scan data are generally noisy and have occluded components and cluttered scenes. Also, the data are often susceptible to changes in environmental factors such as rain, fog, dust, or mist.
3. Learning models do not perform well when there is a significant difference between the training and test data.
4. Supervised learning requires extensive annotated training and testing datasets for various categories of incomplete building elements: currently, the data is limited.

ScanNet [40], S3DIS [41], and ModelNET-40 [42] are 3D point cloud datasets consisting of different classes of building elements that are currently used in studies. However, these data sets represent completed buildings; for construction progress monitoring studies, data sets are required to be acquired from buildings under construction. For material recognition, one such example of an image-based dataset is the Construction Material Library (CML) created in [9] and [37], which consist of more than 3000 images categorised into 20 construction material classes.

There are two approaches to generate the data. The first approach is by generating realistic, rendered elements-based graphic models to create synthetic data [43]. The second approach is to capture and process real-world data with the high computing power of state-of-the-art data acquisition devices.

Table 5 Parametric as-built modelling approaches in the literature

Approaches		[74]	[13]	[30]	[39]	[33]	[75]	[76]	[77]	[57]	[12]	[14]	[8]	[78]	[11]	[79]	[26]	[80]	[81]	[82]	[83]	[27]	[84]	[29]	[85]	[86]	[87]	[34]	[88]	[89]
Heuristic Based	**Geometry**	x		x										x	x	x	x	x	x	x	x	x	x	x	x	x	x		x	x
	Rule	x														x	x	x	x	x	x	x	x	x	x	x	x			
	Relationship	x														x	x	x	x	x								x		
	Model		x						x	x	x	x	x	x	x															
Learning-Based	**Learning – Geometry**	x	x	x	x	x	x																							
	Learning – Appearance	x	x		x			x																						

Table 6 Classification of applied heuristics to obtain specific elements

Heuristics Classification		Elements and Applied Heuristics
Geometry-based Dimensional and associative restrictions that can be applied on point clouds, segmented planes, objects, or volumetric representation.	Shape and dimension	Wall: dimensional constraints[82][29] Wall: floor to roof distance [87] Column: dimensional constraints [89] Floor: altitude of plane [87] [27] Floor: location of centroid [86] Roof: altitude of plane [87] Roof:location of centroid [86] Pipe: dimensional constraints [11] Pipe: circular cross-section [11] Window: rectangular [87] [90] Door: narrow regions along the trajectory of the device [84] Door: rectangle shape constraints [87] [91] [90] Opening: shape [87] [78] Secondary components : shape [78]
	Point density	Floor: point density [89] Roof: point density [89] Window: low point density [87] [82] Wall: 2D countour generation [84]
	Associative geometry	Opening: the location on the wall [87] Beam: plane projection and line fitting [89] Window & door: distance from the wall, ceiling and adjacent wall [87] [90]
Rule-based These are the general heuristics most of the building elements follow.	Surface normal	Floor: surface normal opposite to direction of gravity [87] [27] [80] [81] Roof: surface normal in direction of gravity [87] [80] [81]
	Orientation	Wall: orthogonal to ground [87] [27] Wall: verticallity constraint [82][81] [27][80] Window & door: alignment [90]
	Principal axis	Door: axis constraints [84]
	Other	Window: never stand-alone [82] Door: never stand-alone [82]
Relationship-based Relationship-based heuristics can be time-based or space-based.	Time-based (e.g., the sequence of construction objects)	Sequential relationships [34]
	Space-based (e.g., connections between objects or spaces).	Wall: intersects with ground [82] [34] Roof: on top of walls and intersect on top of walls [82] Window: positioned in walls [82] [34] Door: positioned in walls [82] [34] Opening: the location on the wall [87]
Model-based Uses as-planned BIM model to detect elements using overlap with as-built data.	Derived from existing model	Pipes: alignment and comparision [11] Elements: voxel occupancy [8] Elements: ray thresholding [12] Elements: commercial software [14] Elements: thresholding [57] Elements: machine learning [13] Elements: point intensity [77] Secondary components: RGB comparison [78] Door: RGB comparison [91]

If the challenges mentioned earlier are solved, the learning-based techniques open doors to autonomous element detection and identification for robust progress monitoring.

Hybrid Approaches

As discussed, both the heuristics-based approach and the learning-based approach face a set of challenges. While the former requires extensive hard coding of the domain knowledge, which is complex and challenging to define for all situations faced in construction, the latter approach requires extensive training datasets, higher computational resources, and knowledge of advanced computing techniques. Though heuristics-based approaches may work for general geometric components, they fail to perform when the construction scene becomes complex and do not follow conventional rules, relationships, and geometries. Learning-based approaches are more adaptive, as they can generalise on-site data and are not based on structured domain knowledge.

Therefore, to utilise the advantages of both the approaches and overcome their individual shortcomings, there is a need to explore the hybrid approach, i.e., using heuristics in conjunction with learning algorithms.

3.3. Progress monitoring

Progress monitoring requires a visual or quantitative comparison between the as-planned and as-built models. Four progress monitoring levels identified as a part of this study are L-1, L-2, L-3, and L-4, as discussed in section 2. The following sub-section discusses the key technologies and methods used for implementing these levels.

3.3.1. Progress visualisation and comparison (L-1 and L-2)

L-1 based monitoring is currently the predominant method used at construction sites and only requires visualising the 3D as-built model built from spatial data obtained from the site. As an extension to L-1, L-2 based monitoring requires a visual comparison between as-planned and as-built status at a given time. This is also used on several projects and explored by numerous research studies.

These visualisations or comparisons can be made using selected environments, as shown in Figure 2 (e). The literature review in Table 1 shows that most pipelines have used non-immersive environments such as a 3D viewer or web-based viewer. A few pipelines have used immersive Extended Reality (XR) environments, which include Augmented Reality (AR) [21][28], Virtual Reality (VR) [13][44], and Mixed Reality (MR) [45][12].

While a few studies used traffic light-based coding to show progress [21], the use of comparative and interactive UI has also been investigated [28]. Interactive UIs are gaining popularity, as they can pass information among stakeholders and assign tasks or comments for the workforce by adding features like annotating PDF notes into the environments. Advancements in gaming engines can be used to represent point clouds and BIM models by importing and overlaying in a single VR environment utilising Unity3D [13][44].

Based on the review of literature and practice, this study finds that the capability of visualisation environments is underutilised for progress monitoring applications. With advancements in computing skills, rapid development in immersive technologies, and required bandwidths supported by the 5G standard, these platforms can provide the required information in real-time [46]. Focused research is required to generate interactive UI and integrate AR and VR environments to give the user the best of both worlds and utilise gaming engines to take progress visualisation to the next level.

3.3.2. Progress quantification, schedule updating, and notifications (L-3 and L-4)

The next level of progress monitoring, L-3, involves quantifying the completed elements. Therefore, in continuation to Table 4, Figure 5 shows a trade-off for selecting the quantity estimation method corresponding to the type of model generated. Here, on the left-hand scale, the green-to-red colour variation indicates the increase in complexity from low to high. Similarly, the green-to-red colour variation indicates the decrease in the ease of progress quantification from high to low on the right-hand scale.

While the IFC/BIM models can produce direct BoQs from IFC-based element geometries [31], other models require comparison with the as-planned model. For surface models and mesh models, object detection (based on models overlapping) [13], detection by ray thresholding [12], or enclosed volumetric comparison [47] can be made for estimating quantities of completed elements.

For comparison using voxel occupancy, the as-planned model is voxelised and overlapped with as-built point clouds, after which thresholding is done to detect the occupied voxels [8]. Although it works well for smaller models, it may be computationally costly for larger models; hence, it is not a preferred option. For direct usage of the point cloud, the most straightforward and widely-used method is Scan-vs-BIM, which uses thresholding-based element identification. This method requires an as-planned 3D BIM and has its shortcomings [48], but the as-built model generation step can be skipped, saving computation. Commercial project management software are also used for detecting deviations between as-planned and as-built models that can result in the quantification of contractual scheduling metrics [14][44].

The selection of a model for progress monitoring depends on the project's characteristics (type, budget, duration), the required level of progress monitoring, and other factors. A detailed study of these factors is required to converge onto an appropriate pipeline for progress monitoring based on the project-specific use case.

For L-4 progress monitoring, updating schedules and generating reports/notifications are included with progress quantification. This is challenging; therefore, as shown in Table 1, only a few pipelines have explored this area. Though schedule updating has not been fully automated yet in practice, conceptual schemes have been proposed [49].

Some schemes have utilized the concept of precedence relationship graphs and articulation points, where all objects must be finished before the element linked to the articulation point can be built [50]. Few studies have compared as-planned and as-built BIM for schedule updating [31][51]. As the as-built IFC/BIM model is updated, parametric information has also been utilised to generate the updated schedules automatically [52].

The primary challenge in schedule updating is that schedule LoD is different from BIM LoD, resulting in mismatch and incompatibility. Secondly, it requires an as-planned 4D model for smooth implementation, that is not always available for a construction project. Other challenges include deciding the frequency of updating 4D-BIM. Although a few research papers on CV-CPM have mentioned the concept of schedule updating in their pipelines, they have not specified the details. Therefore, there is a necessity to conduct and integrate focused studies for tackling these specific challenges.

Likewise, only a few studies have explored an automatic notification system that can trigger notifications while monitoring and controlling a construction site [47]. These can be obtained via SMS or e-mail, containing warnings, reports, or graphical outputs based on the settings and LoD required. The frequency and content of notifications should be selected to give construction managers broad insights about areas to target for meeting schedules. As discussed earlier, providing managers with real-time information/alerts about any deviation in the schedules or discrepancies in the as-built model from the as-planned model is essential for in-time decision making and can improve the project's performance.

In addition to providing updated information on progress status, features to suggest interventions and control strategies and forecast resulting outcomes will automate project controls even further.

Conventional control decisions taken by the project managers are based on the know-how of issues specific to the project. Automated control suggestions can be generated using cognitive computing on the progress status and the data from sources such as the organisation's Enterprise Resource Portal (ERP). Such an approach can recognise patterns in activities that suffer delays and suggest interventions to bring the project back on track. There have been selected deployments using AI-based simulation engines for addressing complications inherent in scheduling and project coordination [53]. Such deployments can also evaluate alternatives and provide real-time suggestions. This is an emerging area with significant potential, and further exploration is required.

This section presented a micro-analysis of the stages of the assembled CV-CPM framework. Various gaps were identified while discussing each section in detail. The following section discusses the application of the CV-CPM framework to enable broader evaluation of various pipelines/components to develop benchmarks. It also summarises specific challenges identified within each stage.

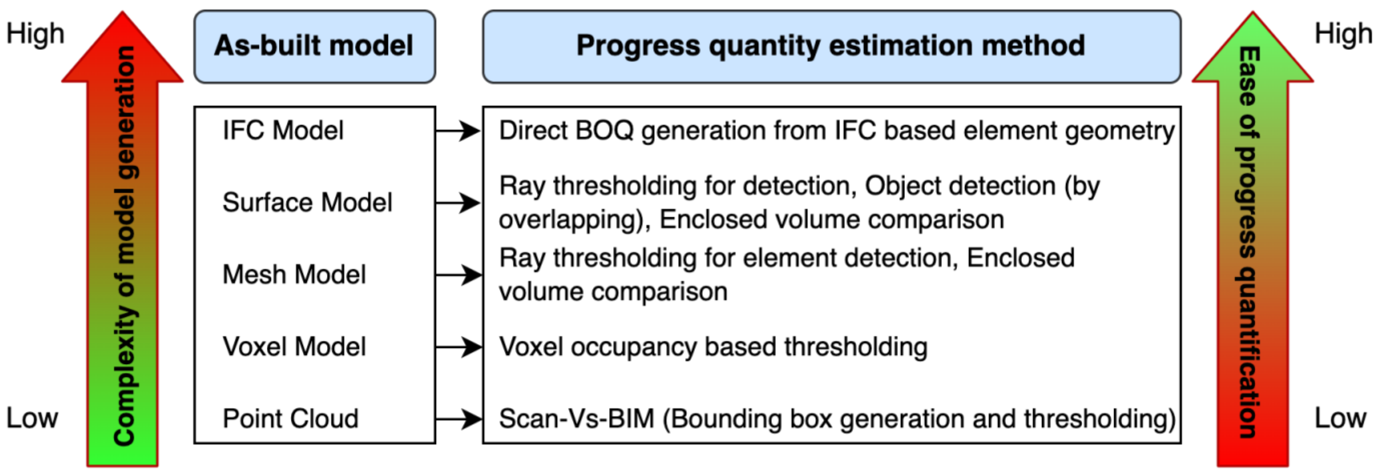

Figure 5 Trade-off for selecting Quantity Estimation method corresponding to the type of as-built model for progress quantification

4. Discussion

As presented and discussed in the previous section, numerous combinations of input-process-output options are possible at each stage of the CV-CPM framework. For an integrated pipeline across all stages, the combinations of options are immense. Given the pace at which technology is developing, the number of options will continue to increase. However, only a limited combination of options will be feasible, and this paper has proposed guidelines to structure these combinations.

A systematic approach is essential to explore and document the performance of specific approaches/algorithms and integrated CV-CPM pipelines. This section discusses requirements for benchmarking to enable a systematic study. The section also summarises potential research questions within each stage of CV-CPM and discusses its relevance to support the emerging area of Digital Twin.

Benchmarking

Benchmarking evaluates pipelines, algorithms, devices, and tools using standard datasets and test methods. Figure 6 outlines the broad requirements to benchmark pipelines and components of CV-CPM, and these are discussed below:

1. Datasets

Though existing datasets contain relevant information for building elements [40][41], these represent completed elements and not the schedule-based construction progress of the elements as required for progress monitoring. Hence, there is a requirement for creating an open-source dataset that contains the data on elements as construction progresses. As shown in Figure 6, these can be synthetically created or acquired from the real world. The dataset type will change based on the particular stage under investigation. For example, for 3D reconstruction, the type of dataset required for benchmarking reconstruction algorithms will be in the form of raw images, videos, and depth images.

2. Testbeds

While existing studies in the area have contributed significantly to knowledge about the applicability of specific algorithms and tools used, the experimental set-ups and procedures used for these investigations vary. As a result, repeatability, and comparison of results across research groups are not feasible. Hence, for each CV-CPM stage, there is a need to design standard testbeds that define the experimental setup, procedures, and measurement parameters. As shown in Figure 6, the testbed can be either computational for benchmarking software (algorithms) or physical for benchmarking the hardware (devices/sensors). For example, a testbed for data acquisition can have sensors capturing data with various physical constraints, like the speed of capture, distance from an object, lightning condition, resolution settings etc., that can be varied to perform multiple experiments. Further, the testbeds should be flexible to accommodate the requirements for collaborative explorations in the area.

3. Component of CV-CPM

As shown in Figure 6, for CV-CPM, the components can be benchmarked independently or as pipelines integrating multiple components. Each component can be evaluated for its independent performance. Additionally, the pipelines can also be benchmarked for their effectiveness for overall progress monitoring by varying specific components within the pipeline. This will standardise the approach to sequence and investigate various input-process-output combinations and enable a systematic comparison of results with other studies.

4. Evaluation Metrics

There are several factors to be considered for evaluation. While most benchmarking studies within this framework require the definition of quantitative metrics, qualitative measures will also be required to evaluate subjective outcomes. For example, for benchmarking visualisation environments, qualitative factors like ease of use, skills required for navigation, immersive features for comparison, and training requirements can be compared from a user feedback-based evaluation. Correspondingly, for benchmarking the algorithms, the evaluation metrics can be in the form of accuracy, processing time, computational cost etc., which are quantitative.

The integrated CV-CPM framework and the guidelines proposed for each stage is expected to assist in developing a strategy and a roadmap for benchmarking. As a wide range of studies is required, the technology roadmap needs to be developed collaboratively by the research community. A starting point for these studies can be to investigate the subjective ratings proposed in this study and quantify these ratings through controlled experimental testbeds. The CV-CPM framework can help in structuring and prioritising the areas to be explored in developing a roadmap so that they form a standard reference for benchmarking studies.

Future research directions

This section highlights potential research questions within the three stages of the CV-CPM framework. Several of these questions can be addressed through benchmarking studies, while others require exploration of new concepts.

Stage 1: Data acquisition and 3D reconstruction

- How can we objectively assess the factors affecting data acquisition technology and sensor mounting method shown in Figure 3?

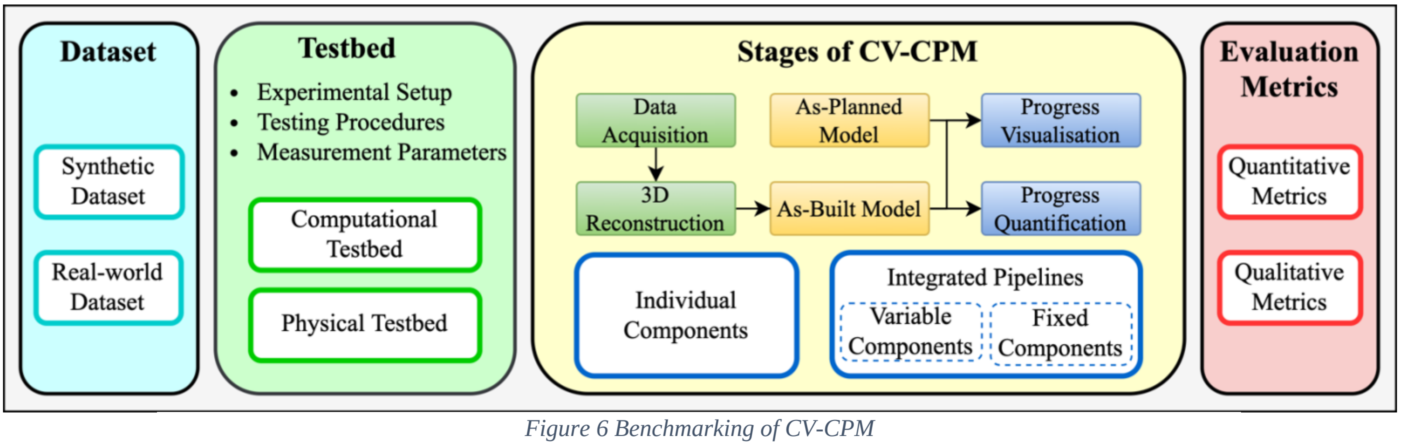

Figure 6 Benchmarking of CV-CPM

- What are the appropriate sensing technology and mounting method for data acquisition based on project types and project characteristics (ex., Linear, underground, elevated etc.)?
- Based on factors identified in Figure 4, How can we decide on using SfM and SLAM for a specific project?
- Vision-based data is usually enormous, and storage requirements are high. How sparse can data be without impacting the output?

Stage 2: As-built modelling
- How can the data be effectively pre-processed to meet the point cloud quality requirements for an as-built model generation?
- How can a hybrid of heuristics and learning-based approaches be deployed, such that it exploits the advantages and overcomes the shortcomings of each?
- How can cognitive computing be successfully implemented to recognise and measure complex geometries at reduced computational cost accurately?

Stage 3: Progress Monitoring
- How can we objectively compare the computational requirements and output accuracy for the level of progress monitoring based on the type of as-built models as shown in Table 4?
- How can virtual and physical environments be seamlessly connected to improve visualisation?
- How can we select an appropriate as-built model and progress quantification technique based on project characteristics and the trade-off mentioned in Figure 5?
- How can we generate automatic schedules update and control suggestions by applying cognitive computing techniques to the progress data produced by CV-CPM?

In addition to the specific areas identified above, the CV-CPM framework is also relevant to support emerging areas of research and development, such as Digital Twin.

Digital Twin technologies enable the built facility's virtual representation to mirror and predict the state/behaviour of the physical facility. Key technologies for Digital Twin implementation include:
1. continuous streaming and processing of sensing data,
2. threaded models and simulations for current/future state/behaviour of the facility,
3. formulating/implementing interventions to limit future state/behaviour within specified limits.

Studies have envisioned the utility of Digital Twin in all phases of a construction project lifecycle. However, details on twinning an under-construction facility are limited [54]. Commercial platforms such as Autodesk Tandem [55] and Bentley iTwin [56] are available for creating Digital Twin solutions. However, technological features available in these platforms are currently in the early stages and focus on the operations and maintenance phase of the built asset.

A key issue in creating a Digital Twin for construction progress monitoring is that the geometry of the facility keeps expanding; hence the corresponding sensor positioning is also dynamic. In this context, computer vision-based sensing would be the most appropriate input to capture geometrical attributes directly and rapidly. The proposed CV-CPM framework identifies the essential technologies to embed progress monitoring capabilities in a Digital Twin.

Pipelines that can stream and process the progress data accurately and rapidly to mirror the progress in the virtual model will support Digital Twinning for progress monitoring. For this, the process within and across the stages of the CV-CPM framework needs to be optimised for real-time updates.

In addition to mirroring the progress, a Digital Twin can forecast construction progress as well as simulate and evaluate control measures to bring the project back on track. As the usage of autonomous construction equipment increases, the control measures can be transmitted from the Digital to the physical world to be implemented by the equipment.

5. Conclusion

CV-CPM has the potential of creating an immense impact by providing real-time, accurate, reliable information to construction managers. Though a significant amount of work has been done in the last decade, specific challenges remain due to the construction industry's dynamic nature and complexities at sites. Some of these challenges have been addressed; however, significant gaps need to be filled to make pipelines accurate and automated to meet rising user expectations of real-time feedback.

This study has comprehensively reviewed individual pipelines and formulated an integrated CV-CPM framework, as shown in Table 1 and Figure 2, respectively. The proposed framework positions all the reviewed papers, including recent papers in this area. Hence, it can be inferred that it is holistic and robust.

It was found that the four levels of progress monitoring identified in this study strongly influence the technology selection for the pipeline at each stage. For implementing progress monitoring on a construction project, it is recommended that the pipeline and components required are selected based on the chosen level.

Several areas for further research were discussed in the previous section. Among these areas, advancements in the as-built modelling stage are required to facilitate the quantification of progress. To enable this, exploring a hybrid approach that combines learning with heuristics is recommended.

The need and requirements for benchmarking and future research directions derived from the CV-CPM framework have been presented in the Discussion section. Addressing these requirements and following a well-developed roadmap in this area is essential to move CV-CPM research from laboratory studies to field applications.

Abbreviations

The following abbreviations are used in this manuscript:

AEC Architecture, Engineering and Construction
AI Artificial Intelligence
AR Augmented Reality
BIM Building Information Modelling
BoQ Bill of Quantities
CAD Computer-Aided Design
CCTV Closed-Circuit Television
CMVS Clustering Multi-View Stereo
CNN Convolutional Neural Network
CV-CPM Computer Vision-based Construction Progress Monitoring
DT Digital Twin
DL Deep Learning
GPS Global Positioning System
GPU Graphical Processing Unit
GUI Graphical User Interface
HSI Hue-Saturation-Intensity
ICP Iterative Closest Point
IFC Industry Foundation Class
IMU Inertial Measurement Unit
IoT Internet of Things
LiDAR Light Detection and Ranging

LoD Level of Detail
LPM Level of Progress Monitoring
MEP Mechanical, Electrical, and Plumbing
ML Machine Learning
MR Mixed Reality
MVS Multi-View Stereo
PCA Principal Component Analysis
PMVS Patch-based Multi-View Stereo
PSR Poisson's Surface Reconstruction
QR Quick Response
RANSAC Random Sample Consensus
RFID Radio Frequency Identification
RGBD Red Green Blue Depth
SfM Structure-from-Motion
SGM Semi-Global Matching
SLAM Simultaneous Localisation and Mapping
SMS Short Message Service
TLS Terrestrial Laser Scanning
UAV Unmanned Aerial Vehicle
UGV Unmanned Ground Vehicle
VR Virtual Reality
XR Extended Reality

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